define the distance

$$d(p,q) = \inf\{L(\gamma) : \gamma \text{ pcwise regular } p \text{ to } q\}$$

Lemma This definition makes (M, g) a metric space, and the metric topology agrees with the manifold topology.

Proof: symmetry, ≥ 0 and triangle ineq. follow immediately from the def.

For the other properties, we need to compare norms.

Let $x^1, \dots x^n$ be normal coord's at p, radius $\geq r$

$$V \to v^i \partial_i$$

We have two norms $|v|_g^2 = g_{ij} v_i v_j$

$|v|_\delta^2 = \delta_{ij} v_i v_j$ (Euclidean)

The function $|\cdot|_g$ is continuous and non-vanishing on the Euclidean opt set

$$K = \{(x,v) : |x| \leq r, |v|_\delta \leq 1\}$$

$\Rightarrow \exists$ constants $c_1, c_2 > 0$ so that

$$c_1 |v|_\delta \leq |v|_g \leq c_2 |v|_\delta \quad (\text{"comparable"})$$

$\Rightarrow$ Within $B_r(p)$, L is comparable to L_δ defined using $\int |\dot\gamma|_\delta \, dt$

$\Rightarrow d_g$ and d_δ are comparable in $B_r(p)$

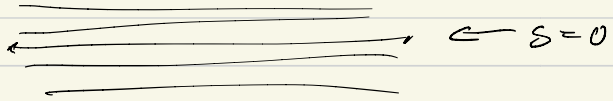
Also for $q \notin B_r(p)$ this shows $d(p,q) \geq c_1 r$

Minimizing length

Consider a family of regular curves

$$F : (-\varepsilon, \varepsilon) \times [a,b] \to M \quad \text{smooth}$$

such that $\Gamma(s,\cdot)$ is regular for each s



(both allows for piecewise regular)

Both $\partial_s F$ and $\partial_t F$ qualify as vector fields along Γ

The variation is ~~proper~~ if endpts. $p = \Gamma(s,0)$ and $q = \Gamma(s,1)$ are fixed

We'll study the length functional $L(s) = l(\Gamma(s,\cdot))$

Symmetry Lemma For F as above

$$D_s \partial_t F = D_t \partial_s F$$

Proof: The derivatives are local so we can use coord's

$$F(s,t) \rightsquigarrow x^i(s,t) \qquad i = 1, \dots n$$

$$\partial_t F = \frac{\partial x^i}{\partial t} \partial_i$$

recall: $D_s(v^i \partial_i) = \left[\frac{\partial v^i}{\partial s} + v^j \frac{\partial x^k}{\partial s} \Gamma^i_{jk} \right] \partial_i$ (avoiding dots here)

$$\to D_s \partial_t F = \left[\frac{\partial}{\partial s}\left(\frac{\partial x^i}{\partial t}\right) + \frac{\partial x^j}{\partial t}\frac{\partial x^k}{\partial s}\Gamma^i_{jk} \right] \partial_i$$

Similarly $$D_t \partial_s F = \left[\frac{\partial}{\partial t}\left(\frac{\partial x^i}{\partial s}\right) + \frac{\partial x^j}{\partial s}\frac{\partial x^k}{\partial t}\Gamma^i_{jk} \right] \partial_i$$

These are equal, since $\Gamma^i_{jk} = \Gamma^i_{kj}$

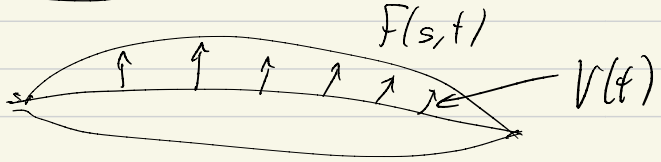
Variational calculus

Given $\gamma:[a,b] \to M$ regular, unit speed, consider a proper regular variation F such that

$$F(0,t) = \gamma(t)$$

Define $L(s) := \ell(F^s(s,\cdot))$

The variation field along γ is $V(t) := \partial_s F(s,t)\big|_{s=0}$



First Variation Formula: For a proper regular variation of unit speed γ,

$$L'(0) = -\int_a^b g(V, D_t\dot\gamma)\,dt$$

Proof: Let $T := \partial_t F(s,t)$, $S = \partial_s F(s,t)$

$$L(s) = \int_a^b \sqrt{g(T,T)}\,dt$$

$$L'(s) = \int_a^b \frac{\partial_s g(T,T)}{2\sqrt{g(T,T)}}\,dt \qquad \partial_s g(T,T) = 2g(D_s T, T)$$

$$= \int_a^b \frac{1}{|T|_g}\cdot g(D_s T, T)\,dt$$

$$= \int_a^b \frac{1}{|T|_g}\, g(D_t S, T)\,dt \qquad \text{symmetry lemma}$$

$$\Longrightarrow L'(0) = \int_a^b g(D_t V, \dot\gamma)\,dt$$

$$\partial_t g(V,\dot\gamma) = g(D_t V, \dot\gamma) + g(V, D_t\dot\gamma)$$

and $$\int_a^b \partial_t g(V,\dot\gamma)\,dt = g(V,\dot\gamma)\Big|_a^b = 0$$

$$L'(0) = -\int_a^b g(V, D_t\dot\gamma)\,dt$$

Note: Given V along γ, we can define a correspondly variation by taking

$$F(s,t) = \exp(sV(t)) \quad \text{for } s \text{ sufficiently small}$$

Thm Suppose γ is unit speed and has minimal length among regular curves p to q. Then γ is geodesic.

Proof: For any variation F, $L(s)$ has a minimum at $s=0 \Longrightarrow L'(0)=0$

Take $\alpha \in C^\infty[a,b]$ with $\alpha > 0$ on (a,b), $\alpha(a)=\alpha(b)=0$

$\Rightarrow V = \alpha D_t\dot{\gamma}$ is a proper variation field along γ

$L'(0)=0$ for this $V \Longrightarrow \int_a^b \alpha |D_t\dot{\gamma}|_g^2 \, dt = 0$

$\alpha > 0$ on $(a,b) \Rightarrow D_t\dot{\gamma} \equiv 0$ on a,b.

($D_t\dot{\gamma}=0$ is the "Euler-Lagrange" equation for the length function)